# Python and NumPy for machine learning

**Today:**

- Basic programming, elementary linear algebra, means, and variances.

**Before we start:**

- Check that you have a working python environment

# Today

**By the end of this session, you can:**

1. write small Python functions with lists, loops, and comprehensions;
2. reason about NumPy **shapes**, **dtypes**, and **axes**;
3. vectorize a calculation instead of writing loops;
4. generate reproducible data with an explicit random-number generator;
5. **fit, predict, and evaluate** a univariate linear-regression model;
6. verify results with invariants and `numpy.linalg.lstsq`.

# Course roadmap

1. Python, NumPy, SciPy, and linear regression
2. Ridge regression with GD, momentum/NAG, and SGD
3. Elastic Net with proximal gradient descent
4. Gaussian mixtures with projected gradient and EM
5. Reproducibility, metrics, and evaluation
6. Final exam

# Assessment

- **Duration:** three hours in class
- **Resources:** access to online documentation
- **Not allowed:** LLM assistance
- **Format:** one exercise closely related to class work

**A typical exam asks you to adapt an optimization algorithm or apply the course methods to a real-world problem.**

# Python essentials

**The syntax needed for today's fit–predict–evaluate workflow.**

# How Python runs

- You run a Python file directly. There is no separate compile step before you see results.
- Statements run in order, top to bottom.
- A function's body runs only when you call it.
- When something fails, read the **last** line of the error first.

💡 **Libraries like NumPy move the heavy numerical work into fast compiled code (details in appendix B)**

# Run a script

**`hello.py`:**

```
message = "Hello, world!"
print(message)
```

**From the repository environment:**

```
uv run python hello.py
```

**Output:**

```
Hello, world!
```

**`uv` handles the Python environment for you. See the *Python tooling & environments* part of the appendix for `uv init`, `uv sync`, and how environments work.**

# Names, values, and types

**The four classic types:**

```
x = 42            # int
pi = 3.14         # float
name = "Alice"    # str
is_valid = True   # bool

print(type(x), type(pi))
```

- A name refers to a value.
- Values have runtime types; incompatible operations can fail at runtime.
- Type hints (`x: int = 42`) document intent for editors; Python does not enforce them.

# Strings

## Combine strings in several ways:

```
greeting = "Hello"
name = "Alice"
msg = f"{greeting}, {name}!"   # f-string: embed values

print(msg)  # Hello, Alice!
```

## Many methods exist:

```
text = "  Python Basics  "
text.strip()          # "Python Basics"
text.lower()          # "  python basics  " (whitespace preserved)
"-".join(["A", "B"])  # "A-B"
```

# Lists: ordered values

```
values = [2.0, 4.0, 8.0, 16.0]

values[0]        # 2.0
values[-1]       # 16.0
values[1:3]      # [4.0, 8.0]
len(values)      # 4
```

**Indexes start at zero. A slice includes its start and excludes its stop.**

```
 index       0    1    2     3
 value     2.0  4.0  8.0  16.0
```

# Lists are mutable

```
values = [2.0, 4.0]
values.append(8.0)
values[0] = 1.0

print(values)  # [1.0, 4.0, 8.0]
```

**Assignment does not copy a list:**

```
alias = values
alias.append(16.0)
print(values)  # [1.0, 4.0, 8.0, 16.0]
```

**Use `values.copy()` for an independent shallow copy.**

# Tuples: immutable, ordered

```
point = (2, 3)
x, y = point      # unpacking

print(x)  # 2
```

- **Immutable**: once created, cannot be changed.
- **Ordered**: like lists, preserve insertion order.
- Support unpacking: `x, y = point`.

**Use a tuple when the size and meaning are fixed, like a point `(x, y)`.**

# Dictionaries map keys to values

```
person = {"name": "Alice", "age": 30}

print(person["name"])  # Alice
```

**Safely read and update:**

```
person.get("age", "unknown")   # 30
person.update({"age": 31})

print(person["age"])  # 31
```

**Use a dict when you need lookup by a label rather than by integer index.**

# Sets hold unique values

```
tags = set([1, 2, 2, 3])
print(tags)  # {1, 2, 3}
```

```
tags.add(4)
tags.union({5, 6})
```

**Unordered iteration; duplicates are automatically removed; supports union, intersection, and difference; mutable.**

# Conditions

```
mse = 2.3
baseline_mse = 51.5

if mse < baseline_mse:
    print("the model beats the baseline")
else:
    print("check the implementation")
```

**Blocks are defined by indentation. Comparisons produce Booleans.**

**Useful operators:** `==`, `!=`, `<`, `<=`, `>`, `>=`, `and`, `or`, `not`.

# Loops and `range`

```
squared = []
for value in range(5):
    squared.append(value**2)

print(squared)  # [0, 1, 4, 9, 16]
```

**`range(5)` produces integers from `0` up to but not including `5`.**

**Use a loop when each iteration performs a meaningful step. Later, NumPy will replace many numerical loops with array operations.**

# Comprehensions express simple transforms

**Loop form:**

```
predictions = []
for value in x_values:
    predictions.append(intercept + slope * value)
```

**Comprehension:**

```
predictions = [
    intercept + slope * value
    for value in x_values
]
```

**Use the comprehension for a simple transform; keep the loop when the body has several steps.**

# Iterate over paired data with `zip`

```
observed = [3.0, 5.0, 8.0]
predicted = [2.5, 5.5, 7.0]

errors = []
for target, prediction in zip(observed, predicted):
    errors.append(target - prediction)

print(errors)  # [0.5, -0.5, 1.0]
```

**`zip` stops at the shortest input. Validate equal lengths when silently dropping data would be an error.**

# Functions package a contract

```python
def mean(values: list[float]) -> float:
    """Return the arithmetic mean of a non-empty list."""
    if not values:
        raise ValueError("mean requires at least one value")
    return sum(values) / len(values)

result = mean([2.0, 4.0, 6.0])
print(result)  # 4.0
```

**A useful numerical function states its inputs, output, and invalid cases.**

# Classes group data with behavior

```python
class Person:
    def __init__(self, name):
        self.name = name

    def say_hello(self):
        print(f"Hey, I'm {self.name}")

p = Person("Alice")
p.say_hello()  # Hey, I'm Alice
```

**We will not need to write classes today, but you will meet them in libraries and later sessions.**

# Imports make dependencies explicit

```
import math

distance = math.sqrt(3.0**2 + 4.0**2)
print(distance)  # 5.0
```

**Later today:**

```
import numpy as np
import matplotlib.pyplot as plt
```

**Keep imports near the start of a script or notebook. Avoid wildcard imports: they hide where names come from.**

# Read the final line of an error first

```
Traceback (most recent call last):
  File "metrics.py", line 8, in <module>
    score = mse_py(targets, predictions)
  File "metrics.py", line 3, in mse_py
    raise ValueError("inputs must have the same non-zero length")
ValueError: inputs must have the same non-zero length
```

1. Read the exception type and message.
2. Find the first frame that belongs to your code.
3. Inspect the values at that line.
4. Fix the cause, not the symptom.

# Bridge to the shared dataset

**The notebook generates one reproducible NumPy dataset, then exposes the same observations as lists:**

```
rng = np.random.default_rng(42)
x = rng.uniform(0.0, 10.0, size=50)
y = 1.0 + 2.5 * x + rng.normal(0.0, 2.0, size=x.shape)

x_list = x.tolist()
y_list = y.tolist()
```

**The Python and NumPy implementations therefore solve exactly the same problem.**

# Exercise · pure-Python MSE

**Time: 7 minutes coding · 15 minutes including launch and debrief · Work: pairs**

**Implement:**

```
def mse_py(y_true: list[float], y_pred: list[float]) -> float:
    """Return the mean squared error of equal-length non-empty lists."""
    ...
```

**Checks:**

```
assert mse_py([1.0, 2.0], [1.0, 2.0]) == 0.0
assert mse_py([1.0, 3.0], [2.0, 1.0]) == 2.5
```

**Reject empty or mismatched inputs with `ValueError`.**

# Exercise hint

**After three minutes:**

1. check the two lengths;
2. pair targets and predictions with `zip`;
3. square each difference;
4. divide the total by the number of targets.

```
squared_errors = [
    (target - prediction) ** 2
    for target, prediction in zip(y_true, y_pred)
]
```

**What should happen if either input is empty?**

# One solution

```python
def mse_py(y_true: list[float], y_pred: list[float]) -> float:
    if not y_true or len(y_true) != len(y_pred):
        raise ValueError("inputs must have the same non-zero length")
    return sum(
        (target - prediction) ** 2
        for target, prediction in zip(y_true, y_pred)
    ) / len(y_true)
```

**The checks are examples, not a complete test suite.**

# Python checklist for today

**You have met:**

- run a script and read an error bottom-up;
- the classic types, strings, lists, tuples, dicts, and sets;
- conditions, loops, comprehensions, and `zip`;
- functions with explicit contracts and a brief look at classes.

**We keep only what today's fit–predict–evaluate workflow needs; the rest is reference material.**

# NumPy shape reasoning 🔢

**Arrays, axes, broadcasting, and checks for reliable numerical code.**

# An array has a contract

```
import numpy as np

values = np.array([2.0, 4.0, 8.0, 16.0])
```

**Every array has:**

```
values.shape   # (4,)
values.ndim    # 1
values.dtype   # dtype('float64')
values.size    # 4
```

**Before any calculation, ask: What shape do I have? What shape do I need?**

# Create data with an explicit generator

```
rng = np.random.default_rng(42)

x = rng.uniform(0.0, 10.0, size=50)
noise = rng.normal(0.0, 2.0, size=x.shape)
y = 1.0 + 2.5 * x + noise
```

```
x.shape == y.shape == (50,)  # True
```

- One owned generator makes state and reproducibility visible.
- Calling `default_rng(42)` again restarts the same sequence.
- Use `rng.integers(0, 10, size=5)`, not a nonexistent module-level `np.random.integers(...)`.

# One dimension is not a row or column matrix

```
vector = np.array([10.0, 20.0, 30.0])
row = vector[None, :]
column = vector[:, None]

vector.shape   # (3,)
row.shape      # (1, 3)
column.shape   # (3, 1)
```

**These shapes carry different broadcasting and matrix-multiplication behavior.**

**A trailing comma matters: `(3,)` is a one-element tuple.**

# Indexing can remove or preserve an axis

```
matrix = np.array([
    [1.0, 2.0],
    [3.0, 4.0],
    [5.0, 6.0],
])

matrix[0].shape        # (2,)
matrix[0:1].shape      # (1, 2)
matrix[:, 0].shape     # (3,)
matrix[:, 0:1].shape   # (3, 1)
```

**Integer indexing removes an axis. A length-one slice preserves it.**

# Views and copies depend on the operation

**Basic slicing usually shares data:**

```
view = matrix[:, :1]
np.shares_memory(matrix, view)  # True
```

**Advanced indexing usually copies:**

```
copy = matrix[[0, 2]]
np.shares_memory(matrix, copy)  # False
```

**`reshape` and `ravel` return views when possible, but may copy. Use `.copy()` when independence is required.**

# Array arithmetic enables vectorization

```
x = np.array([1.0, 2.0, 3.0])

x + 10.0  # array([11., 12., 13.])
x * x     # array([1., 4., 9.])
x**2      # array([1., 4., 9.])
```

**A whole error calculation:**

```
squared_errors = (y - predictions) ** 2
mse = squared_errors.mean()
```

**`*` is element-wise. Use `@` for matrix multiplication. Array expressions state the calculation and let NumPy run compiled numerical kernels.**

# Reductions remove an axis

**For `matrix.shape == (3, 2)`:**

```
matrix.mean()          # scalar
matrix.mean(axis=0)    # shape (2,): one mean per column
matrix.mean(axis=1)    # shape (3,): one mean per row
```

**Think: the named axis is consumed.**

```
axis=0 ↓ combines rows      axis=1 → combines columns
```

**Always connect an `axis` argument to the input and output shapes.**

# `keepdims=True` preserves a length-one axis

```
row_means = matrix.mean(axis=1, keepdims=True)

matrix.shape      # (3, 2)
row_means.shape   # (3, 1)
```

**Now subtraction is unambiguous:**

```
centered = matrix - row_means
centered.shape  # (3, 2)
```

**Without `keepdims=True`, `(3, 2) - (3,)` is not broadcastable.**

# Broadcasting aligns trailing dimensions

**Two dimensions are compatible when they are equal or one of them is `1`.**

```
(3, 2)
(3, 1)
------
(3, 2)  ✓
```

```
(4, 1, 2)
(   3, 2)
---------
(4, 3, 2)  ✓
```

**Broadcasting behaves as if length-one dimensions were repeated, usually without materializing those copies.**

# Shape check · predict before running

**For `points.shape == (3, 2)`, what are the shapes?**

```
points[0]
points[0:1]
points.mean(axis=0)
points.mean(axis=1, keepdims=True)
points[:, None, :]
points[None, :, :]
```

**Answers:**

```
(2,)  (1, 2)  (2,)  (3, 1)  (3, 1, 2)  (1, 3, 2)
```

**Which pair can broadcast to `(3, 3, 2)`?**

# Exercise · center every row

**Time: 7 minutes coding · 15 minutes including launch and debrief · Work: pairs**

```
points = np.array([
    [0.0, 0.0],
    [1.0, 0.0],
    [1.0, 2.0],
])
```

**Create `row_means` with shape `(3, 1)`, then create `centered` with shape `(3, 2)`, all without a Python loop.**

```
assert row_means.shape == (3, 1)
assert centered.shape == points.shape
assert np.allclose(centered.mean(axis=1), 0.0)
```

# Exercise hint

**After three minutes:**

1. Which axis contains the two coordinates of one row?
2. Which axis must the reduction consume?
3. Which length-one axis must remain for subtraction?

```
row_means = points.mean(axis=..., keepdims=...)
centered = points - row_means
```

**Optional fast finisher: use `points[:, None, :] - points[None, :, :]` and predict the resulting shape.**

# Solution · row centering

```
row_means = points.mean(axis=1, keepdims=True)
centered = points - row_means

assert row_means.shape == (3, 1)
assert centered.shape == points.shape
assert np.allclose(centered.mean(axis=1), 0.0)
```

- `axis=1` consumes the two-coordinate feature axis.
- `keepdims=True` retains a length-one axis, so subtraction broadcasts.
- The checks make the intended shapes and numerical invariant executable.

**Optional fast finisher: continue with pairwise squared distances in the notebook only after these core checks pass.**

# Numerical checks are part of the algorithm

```
assert predictions.shape == y.shape
assert np.isfinite(predictions).all()
assert np.allclose(actual, expected, rtol=1e-12, atol=1e-12)
```

- Use `np.allclose`, not exact equality, for most floating-point results.
- Choose tolerances from the scale and precision of the problem.
- Make shape and finiteness failures immediate.
- Avoid an explicit matrix inverse when a solver such as `np.linalg.lstsq` expresses the problem directly.

# NumPy checklist for today

**You can now:**

- inspect `shape`, `ndim`, `dtype`, and `size`;
- preserve or remove axes intentionally;
- distinguish common views from copies;
- replace numerical loops with array expressions;
- explain reductions using input and output shapes;
- apply the trailing-dimension broadcasting rules;
- verify numerical results with shape, finiteness, and tolerance checks.

**Next: use these tools to fit, predict, and evaluate linear regression.**

# Linear regression · one complete workflow 📈

**Model → fit → predict → evaluate → verify.**

# Problem, data, and model

**One reproducible dataset:**

```
import numpy as np

rng = np.random.default_rng(42)
x = rng.uniform(0.0, 10.0, size=50)
y = 1.0 + 2.5 * x + rng.normal(0.0, 2.0, size=x.shape)
```

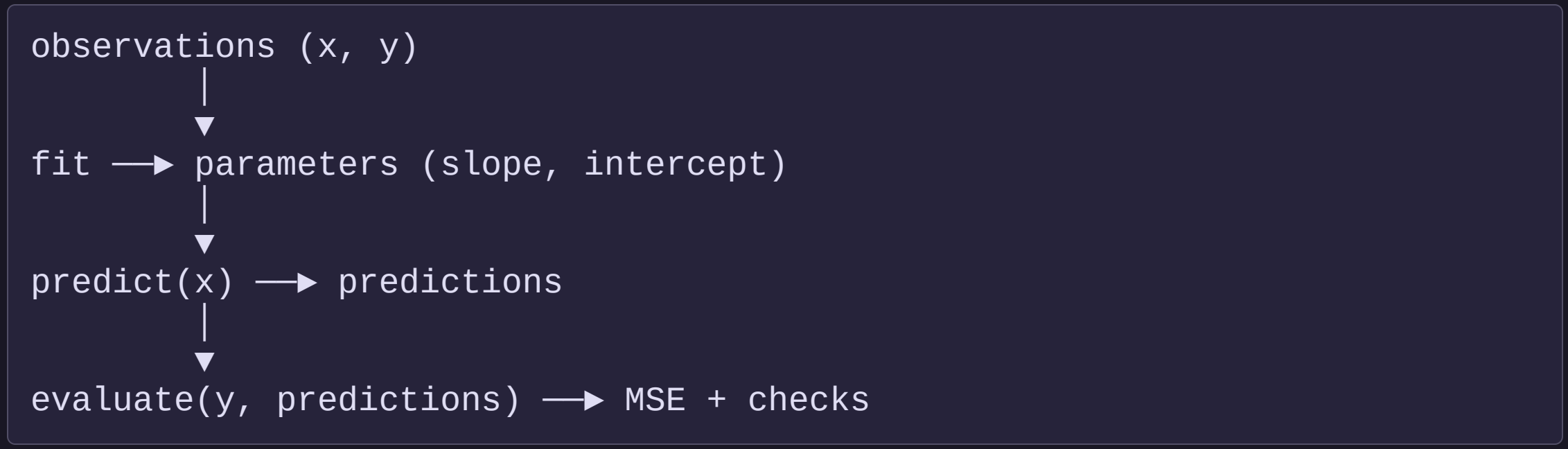
**Model one target from one feature:**

$$\hat{y}_i = b + a x_i$$

**Parameters: slope $a$, intercept $b$; residual: $e_i = y_i - \hat{y}_i$.**

$$\min_{a,b} \ \mathrm{MSE} = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - (b + a x_i)\right)^2$$

# Fit → predict → evaluate




```
baseline = np.full_like(y, y.mean())  # mean-only baseline to compare against
```

**Training MSE checks the implementation, not generalization (that needs held-out data).**

# Prepared baseline · pure-Python fit

**Instructor walkthrough: solution notebook §1. Population factors cancel:**

```
def fit_linear_py(x_values, y_values):
    if not x_values or len(x_values) != len(y_values):
        raise ValueError("x and y need equal non-zero lengths")
    x_mean = mean_py(x_values)
    y_mean = mean_py(y_values)
    numerator = sum(
        (x_i - x_mean) * (y_i - y_mean)
        for x_i, y_i in zip(x_values, y_values)
    )
    denominator = sum((x_i - x_mean) ** 2 for x_i in x_values)
    if denominator == 0.0:
        raise ValueError("x must vary")
    slope = numerator / denominator
    intercept = y_mean - slope * x_mean
    return slope, intercept
```

# Exercise · NumPy fit–predict–evaluate

**Time: 30 minutes · Work: pairs · Implement in `session01_starter.ipynb`:**

```
fit_linear_numpy(x, y) -> (slope, intercept)
predict_numpy(x, slope, intercept) -> predictions
mse_numpy(y, predictions) -> mse
```

**Completion checks:**

```
assert predictions_np.shape == y.shape == (50,)
assert np.isfinite([slope_np, intercept_np, mse_value_np]).all()
assert mse_value_np < mse_numpy(y, np.full_like(y, y.mean()))
```

**Expected: slope ≈ `2.5242`, intercept ≈ `0.5373`, MSE ≈ `2.2939`.**

**Also test invalid inputs: empty, mismatched, 2-D, or constant `x`.**

# Staged hints

## After 10 minutes: shapes

```
x_centered = x - x.mean()
y_centered = y - y.mean()
```

## After 20 minutes: parameters

```
slope_np = np.mean(x_centered * y_centered) / np.mean(x_centered**2)
intercept_np = y.mean() - slope_np * x.mean()
```

## Final check

**The model MSE should beat the mean baseline MSE, approximately `51.4769`.**

# Vectorized solution

```
def fit_linear_numpy(x, y):
    x = np.asarray(x, dtype=np.float64)
    y = np.asarray(y, dtype=np.float64)
    if x.ndim != 1 or y.ndim != 1 or x.shape != y.shape or x.size == 0:
        raise ValueError("x and y need equal non-zero 1D shapes")
    x_centered = x - x.mean()
    denominator = np.mean(x_centered**2)
    if denominator == 0.0:
        raise ValueError("x must vary")
    slope = np.mean(x_centered * (y - y.mean())) / denominator
    intercept = y.mean() - slope * x.mean()
    return float(slope), float(intercept)

slope_np, intercept_np = fit_linear_numpy(x, y)
predictions_np = intercept_np + slope_np * x
mse_value_np = np.mean((y - predictions_np) ** 2)
# (slope, intercept, MSE) ≈ (2.5242, 0.5373, 2.2939)
```

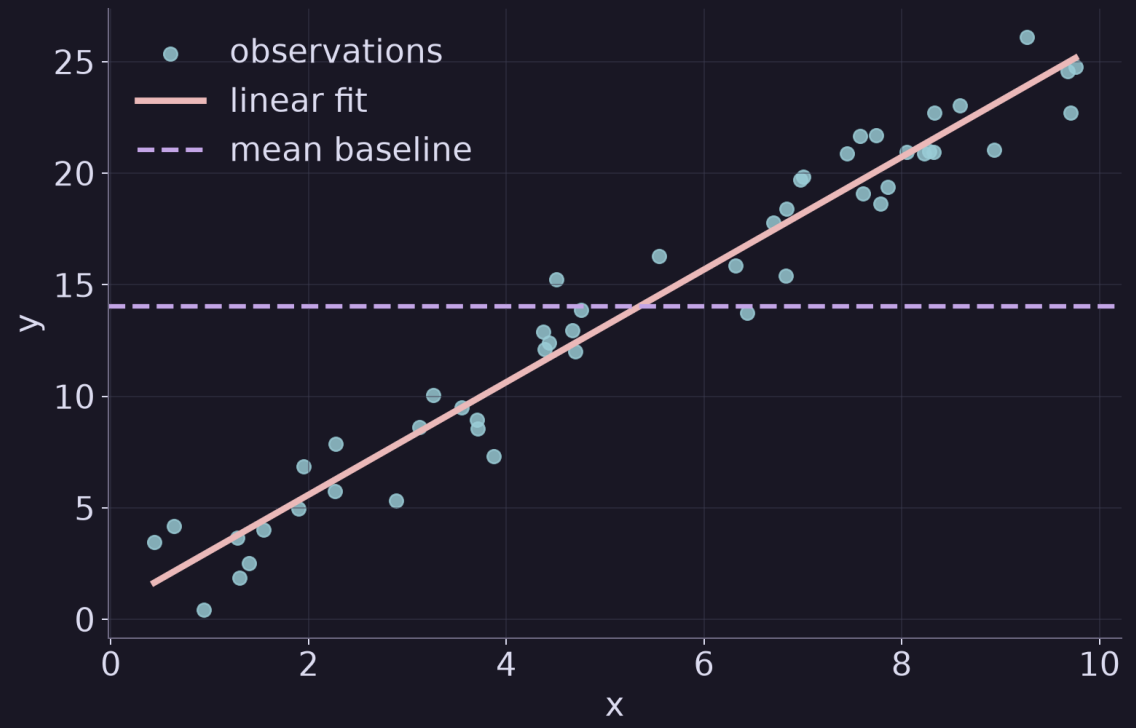
# Verify: solver and plot

```
design = np.column_stack([x, np.ones_like(x)])
(slope_ls, intercept_ls), *_ = np.linalg.lstsq(
    design, y, rcond=None
)

assert design.shape == (50, 2)
assert np.allclose(
    [slope_np, intercept_np], [slope_ls, intercept_ls],
    rtol=1e-12, atol=1e-12,
)
```

`lstsq` **solves the least-squares problem directly. Do not form an explicit inverse just to obtain these coefficients.**

**A plot checks structure; numerical assertions check values.**



# Mini-assessment · 10 minutes

**Answer individually before discussing:**

1. Complete `def predict(x, slope, intercept): return ...`; for `x.shape == (50,)`, what is the output shape?
2. What is the shape of `column_stack([x, ones_like(x)])`, and why include ones?
3. What shape results from subtracting `(50, 1)` predictions from `(50,)` targets?
4. Why use `default_rng(42)`, and why compare MSE with a mean baseline?
5. Does a low training MSE prove good performance on unseen data?
6. What should agree with `lstsq`, and name one invariant to check first.

**Be ready to justify each answer using shapes or the workflow contract.**

# Answers and takeaways

1. `return intercept + slope * x`; output shape `(50,)`.
2. Design shape `(50, 2)`; the ones multiply the intercept.
3. Broadcasting produces `(50, 50)`, a silent and serious bug.
4. `default_rng(42)` reproduces data; the model must beat a constant.
5. No. Generalization needs data not used for fitting.
6. Coefficients match `lstsq`; check shapes and finiteness first.

```
reproduce → validate → fit → predict → compare → verify/plot → interpret
```

**Next: held-out validation, then optimization and regularization.**

# Appendix · reference material

**Not part of the 180-minute lesson. Use these slides after class or when a question comes up.**

- Python background: history, bytecode and the GIL, interpreted vs compiled code
- Resources for further reading
- Python tooling & environments: running code, environments, and `uv`

# Appendix A · A short history of Python

**Reference only, not part of the 180-minute lesson.**

- **1991**: first public release by Guido van Rossum, for readable general-purpose programming.
- **2000**: Python 2.0 (list comprehensions, cleaner stdlib).
- **2008**: Python 3.0, a deliberate breaking redesign; Python 2 retired in 2020.
- **2023–2025**: Python 3.11 to 3.13, with faster startup, richer typing, and better concurrency.
- Main implementation is **CPython** (written in C); neighbors include PyPy (JIT) and MicroPython.

# Appendix B · Bytecode, the GIL, and why NumPy is fast

- **Bytecode**: CPython compiles your source to bytecode, then a virtual machine runs it. You usually never see it; you just run the file.
- **GIL** (Global Interpreter Lock): only one thread executes Python bytecode at a time. It matters only when parallelizing CPU-bound *pure-Python* loops with threads.
- **Why NumPy is fast**: the hot loop is moved into compiled, vectorized C (and Fortran) code, so per-element Python overhead disappears.
- **Going further**: Numba JIT-compiles numeric loops; `pybind11` exposes C code to Python.

**Rule of thumb: keep hot loops in compiled kernels (NumPy, Numba, or a binding), not in pure Python.**

# Appendix C · Interpreted vs compiled

**Reference only.**

**Compiled languages (C, C++, Rust, …) are translated to machine code before running. They are fast at runtime, but you recompile after each change.**

**Interpreted languages (Python, JavaScript, R, …) are run directly by an interpreter or virtual machine. They are easier to test and debug, there is no manual compile step, but raw loops are usually slower.**

**Python trades a little raw speed for much shorter, easier-to-read code. That is exactly why we delegate the hot loops to libraries like NumPy.**

# Appendix C · The dot product in Python and C++

**Python's dot product:**

```python
x = [1, 2, 3]
y = [4, 5, 6]

dot = sum(x_i * y_i for x_i, y_i in zip(x, y))
print(dot)  # 32
```

```
python my_script.py
```

**The same dot in C++ needs a separate compile step first:**

```
g++ dot.cpp -o dot && ./dot
```

# Bonus · setting up `uv`

**Environment setup (`uv init`, `uv sync`, `uv run`, lockfiles, and the full toolbox) is covered in the Python tooling & environments part that follows. The short version for this course: `uv sync`, then always `uv run …`.**

# Resources

**Use when stuck, not as a cover-to-cover read:**

- Python home
- Python documentation
- Python tutorial
- Python standard library reference

# Python tooling & environments

**A companion reference on running, managing, and shipping Python. Separate from the 3-hour lesson and not timed.**

- running code: scripts vs notebooks vs REPL
- which Python is installed, and where
- environments: why and how (`venv`, `uv`, `conda`, …)
- a quick `uv` workflow for this course

# CPython in one paragraph

**The most common Python is CPython: an interpreter written in C that compiles your source to *bytecode* and runs it on a virtual machine.**

- You usually never see bytecode; you just run the `.py` file.
- Other implementations exist: **PyPy** (JIT-compiles hot loops, often faster), **MicroPython** (small embedded devices), **RustPython** (written in Rust).
- "Python" almost always means CPython.

# Three ways to run code

| Way | Command / tool | Best for |
|---|---|---|
| **Script** | `python hello.py` | a whole program end to end |
| **Notebook** | Jupyter `.ipynb` | exploring, mixing text + code, fast feedback |
| **REPL** | `python` (interactive session) | tiny experiments, checking one expression |

**For short questions use the REPL. For a finished pipeline use a script. For today's guided practice use the notebooks.**

# Scripts vs notebooks

**A script runs top to bottom; its state resets each run. Great for reproducibility and automation.**

**A notebook holds cells you run in any order, and its outputs persist. Great for exploration, but order-dependent state can bite you:**

```
cell order is not guaranteed to match "top to bottom"
```

**Rule: explore in a notebook, commit a script for anything you re-run or hand to someone else.**

# Which Python am I actually using?

**Find the interpreter and its version:**

```
which python          # path of the python on your PATH
python --version      # e.g. Python 3.12.11
```

**Inside a virtual environment these point at the environment's interpreter instead of the system one. That is exactly the point.**

# The environment problem

**Different projects need different versions of the same packages.**

- Project A: `numpy 1.26`
- Project B: `numpy 2.0`

**Installing both "globally" is how you break your own machine. The fix is isolated environments: each project gets its own interpreter + dependency set.**

**A lock file (`uv.lock`, `poetry.lock`, `requirements.txt`) records the exact versions so anyone can rebuild the same environment.**

# The environment toolbox at a glance

| Tool | Role |
|---|---|
| **venv** | built-in, minimal virtual environment (`python -m venv`) |
| **pip** | installs packages from PyPI into the current environment |
| **uv** | fast `venv` + `pip` + version/lockfile in one tool (used here) |
| **pyenv** | switch between multiple Python **versions** |
| **conda / mamba / pixi** | environments + compiled/scientific packages, beyond Python only |
| **Poetry / PDM** | full project manager: deps + build + lockfile |

# `venv`, the built-in baseline

```
python -m venv .venv       # create an isolated environment in .venv
source .venv/bin/activate # activate it (Windows: .venv\Scripts\activate)
pip install numpy          # installs only inside this environment
```

**Common pitfall: forgetting to activate, so `pip` installs to the *system* interpreter. `uv` avoids most of this by defaulting to the project environment.**

# `uv`, one fast tool with a lockfile

**`uv` manages the environment, resolves versions, and records a `uv.lock`.**

```
uv sync                    # create env + install pinned deps from uv.lock
uv add numpy               # add a dep, update uv.lock
uv run python hello.py     # run in the project env regardless of activation
uv run jupyter lab         # start the teaching notebooks
```

**Reproducible and fast; this is the workflow this repository uses.**

# `uv init`, create a new project

```
uv init my_project
```

**This creates a standard layout:**

```
my_project/
├── pyproject.toml      # metadata + direct dependencies
├── .python-version     # the Python version for this project
├── main.py             # a "Hello, world!" example entry point
├── README.md
└── .gitignore
```

**It also initializes a git repo (`.git/`) and a `uv.lock` appears on the first `uv sync`.**

## `uv python` and `uv add`, set up the project

**Pick a Python version and add libraries:**

```
uv python install 3.12
uv add numpy
```

**`uv add` records the dependency in `pyproject.toml` and resolves `uv.lock`.**

# Project files for reproducibility

**Inside the project you will find:**

```
pyproject.toml   # metadata + direct dependencies
uv.lock          # exact resolved versions (commit this)
.python-version  # the Python version (3.12 for this course)
```

**Anyone can run `uv sync` from a clean clone and get the same environment. That is what "reproducible" means.**

# Pinning: why `.python-version` and `uv.lock` matter

**Without pinned versions, "it works on my machine" is a coin flip.**

```
# wrong: whatever is installed
uv run python script.py

# right: the exact locked environment
uv sync --locked   # fail if uv.lock would change
uv run python script.py
```

**Reproducibility = deterministic data, pinned languages, pinned tools, pinned versions.**

## `which python`, `uv`, and PATH in one line of mental model

- `which python` tells you the first match on `PATH`.
- A virtualenv puts `.venv/bin` (or `Scripts`) ahead of the system interpreter.
- `uv run` uses the project interpreter without you activating anything.

**So the golden habit for this course: `uv sync` once, then always `uv run …`. No manual activation needed.**

# Resources

- `uv` documentation
- Python packaging guide
- "Hitchhiker's Guide to Python" – environments
- Real Python: virtual environments primer

**Skim when needed; do not read like a book.**